Date 10/10/23

UNIT ⇒ 1

Built Enviroment

- Build Enviroment :- It is everything human created, modified or constructed, human made, arranged or maintained it is the creation of human mind & result of human purpose. It is maintained to serve human needs, wants & Value

- It is created to help us deal with, & to protect from, overall enviroment to mediate or change this enviroment for our comfort & well being.

- Each & all the individual elements contribute either positively or negatively to the overall quality of enviroment both build & natural & to human-enviroment Relationship.

- The term built-enviroment refers to the human made surroundings that provide the setting for human activity, ranging in scale from building & park or ~~gre~~ Green space to neibhourhood & citied than can often inlude their supportive infrastructures. Such as water supply or energy networks.

- The built enviroment is a material & cultural product of human labour that combine physical elements & energy in form of living, working & playing.

- It has been defined as the human made space in which people live, work, & play & recreate on day to day basis.

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→ **Human needs** ⇒ To survive, all organism must satisfy certain basic needs. Human are no exception the most basic set of needs, are physiological - Those required for proper functioning of the body & mind.

→ There are two types of needs.

1. Physicological needs
2. Physicological & social needs.

① **Physicological Needs :-**

(i) Subsistance → Water, Air, Food, Economics, shelter, Technology

compulsory

(ii) **Reproduction** ⇒ Human cycle & Marriage is a compulsory part of human life.

(iii) **Safety** ⇒ Body Protection, shelter, medical, police, fire-protection, weapons, law & governance.

(iv) **Protection from social dysfunction of Insult** ⇒ Ethics, sense of Community, Mental health services.

(v) **Protection from Anxiety & the need to belong** :- The Art, Religion, Cosmology, philosophy, myth & magic.

(vi) **Self Realization** ⇒ Freedom, Responsibility, Education, The Arts

- Human Needs can be physicological or social & are related to security, Respect & self Realization people wants their built environment to be aesthatically Attractive & to be in an accessible place with a well-developed infrastructure, convenent. Communication access & good roads & the develling should also be comparatively cheap, comfortable with low maintainance cost & have sound & thermal Insulation of walls.

→ **Built Environment : purpose :→** The built environment touches all the aspects of our lives, ecompassing the buildings we live in, the distribution system that provide us with water & electricity & the roads, bridge & transportation systems we use to get from place to place. It can generally be described as the man-made or modified structure that provide people with living, working & recreational spaces. creating all these spaces & systems require enormous quantity of materials.

→ **Components of Built Environment :→**

① **Products :–**
- Human Capacity
- Activity easy (Smooth)

- → Machines
  - → Radio
  - → Calculator
  - → Automobiles
  - → Aircraft
- → Graphite Symbol → Western Alphabets
- → Tools
  - → Pen
  - → Pencils
  - → Hammer
  - → Screwdriver
  - → Weapons
- → Materials
  - → Brick
  - → Stone
  - → Wood
  - → Polymers
  - → Plastic

(i) ~~Products:-~~

(ii) Interiors:- Arranged products in an enclosed structure.

- Living Room
- Work Room
- Private Room
- Public assembly halls.
- Stadiums.

(iii) Structures:- are planned grouping of spaces defined by & constructed of products: generally related activities are combined into composite structures (housing, schools, office bldgs, churches, Temples, factories, highways, tunnels, bridges, dams etc.). Structures have both an internal space & an external space.

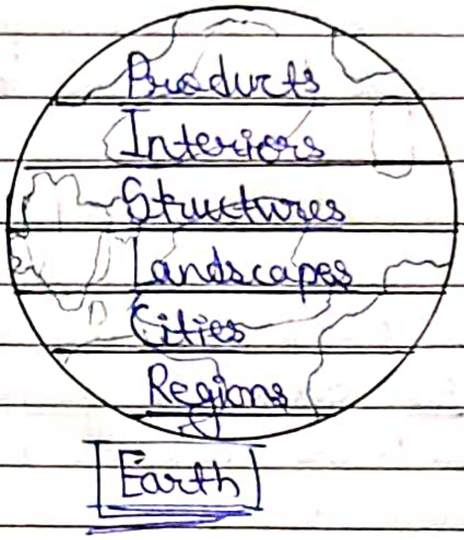
(iv) Landspaces:- are exterior areas and/or settings for planned groupings of spaces & structures (courtyards, malls, parks; gardens, sites for homes or other structures; farms, countryside, national forests & parks). Landscapes generally combine both natural & built environments.

(v) Cities ⇒ Cities are grouping of structures & landscapes of varying sizes & complexities, generally clustered together to define a community for economic, social, cultural, and/or environmental reasons (subdivisions, neighbourhoods, districts, villages, towns, & cities of varying sizes)

(vi) Regions ⇒ are groupings of cities & landscapes of various sizes and complexities; they are generally defined by common political, social, economic, and/or environmental characteristics [the surrounding, regions of cities, countries or multicountry areas (continents), a state or multistate (District) etc].

(vii) Earth ⇒ includes all of above, the groupings of region consisting of cities and landscapes - the entire planet, the spectacular, complex beautiful, still mysterious earth, which as human power expands, may be considered the ultimate artifact. These components can be better understood as connected layers of ~~the~~ levels of varying scales interwoven together to form the built environment.

* Earth ⇒ Region + Interiors + Structures + Landscapes + Cities + Regions + Products



Earth
↓
Products → Regions → Cities
↓
Structures ← Landscapes
↓
Interiors → Products

Elements of Built Environment :- The term built Environment refers to aspects of our surroundings that are built by humans. It includes not only buildings, but the human-made spaces b/w buildings. Such as parks, & the infrastructure that ~~soff~~ support human activity.

- Such as transportation networks, utilities networks, flood defences, telecommunications &

→ Elements of Building ⇒ The following are basic elements of building:

(i) Foundation ⇒ The part of the building, which is usually below the ground & transfers the load of the bldg to the subsoil is known as foundation of bldg.

(ii) Plinth ⇒ The portion of the bldg b/w the surface level ~~of~~ the ground floor is known as plinth.

~~(iii) Walls = The main function of the bldg b/w the surface level of~~

(iii) Walls = The main function of the walls is to enclose & divide the space into various rooms & compartments as per the requirments of the building.

(IV) Columns = The columns are the bldg components which transfer the load of the bldg parts above to it to the plinth & foundation. These are usually provided to take the loads of the beams roofs etc.

(V) Floors = Floors are the components of a bldg over which occupants of the bldg live, move & keep their dwelling material as per their desire. The floor just above the ground is known as ~~gr~~ ground floor. The upper floors above the ground floor are known as first floor, second floor, third floor etc.

(VI) Doors, Windows & Ventilators = These are components provided in the wall for various purposes. Doors serve as a connecting link b/w the internal rooms & also serve as a mean for the movements from inside to outside the bldg.

- Windows are provided for the purpose of light & air inside the bldgs.
- Ventilators are usally meant for taking out foul or hot air from the bldg. Properly ventilated bldg are more cool & comfortable.

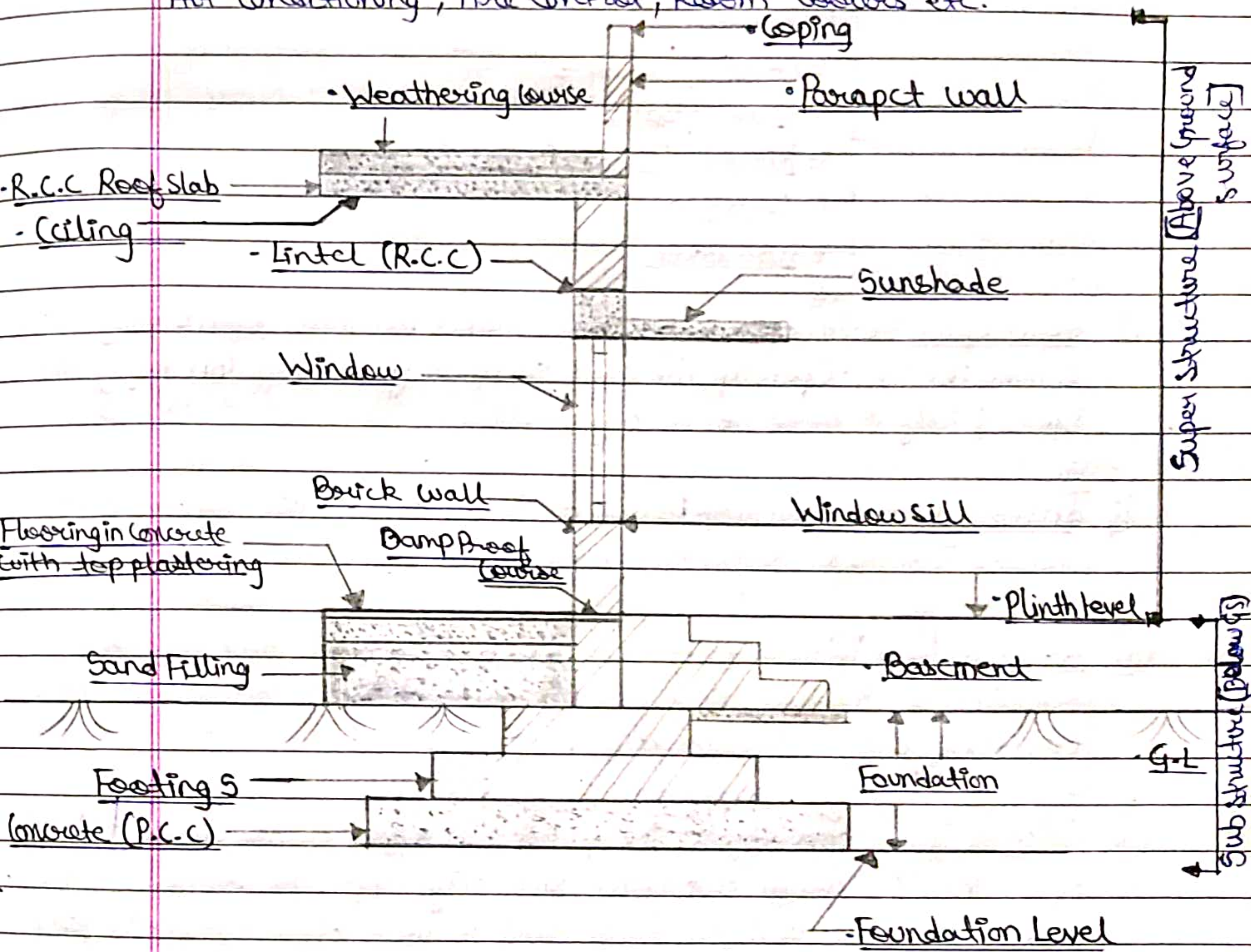
(VII) Stairs = A stair may be defined as a structure comprising of a number of steps ~~for~~ connecting one floor to another. The stair must be constructed in such a manner that it should be so located as to permit easy communication.

(VIII) Roofs = The roof is the upper most horizontal or inclined part of a bldg. The main function being to cover the rooms or enclosers made by walls & to protect it from sun, rain, snow, wind etc.

(ix) Building Finishes ⇒ Bldg finishes include items like Plastering, Pointing white/color washing, Painting, varnishing, distempering, etc.

- The bldg finishes not only perfect the surface from adverse effects of weather but also provide decorative effect.

(x) Building Service ⇒ Bldg service include;: service like water supply, Drainage, Sanitation, Electricity, Acoustics, Heating, Ventilation, Air conditioning, Fire control, Room coolers etc.



Main Components Of A Building

Note:- Above part from Ground Surface is called Super Structure ex(wall, floor, window, lintel, Coping, Parapet wall, Beam, column etc)

Note ⇒ (ii) Below part from Ground Surface is called sub-structure. ex [Basement, foundation, footing, P.C.C (Plain Cement Concrete)] etc.

Office / Commercial Bldg:-

- Open Space
- Internet Connectivity — Telephone, Internet
- Electricity
- Color
- Heating / Cooling
- Parking space
- Kitchen / Dinning Space
- Location
- Furniture

(i) Open Space ⇒ An open office space makes for more open & easy communication. Open up the area so you & your staff can enjoy the space freely & move about any problem.

(ii) Internet Connectivity ⇒ As you all know about this connectivity its include Internet, telephone, broadband, Wifi, router etc.

(iii) Electricity ⇒ Without electricity you dont work in this place becz electricity is the only thing that you want more its include, fan, AC, Electricity board, Switches etc.

(iv) Color ⇒ Color can effect the profound effect on the brain & dull office interior design and color isn't only applicable to your walls even office table and all things that in your office affect the mind. Colourful furniture chairs, tables to add some office.

(v) Heating / Cooling ⇒ Ensuring you have a comfortable climate in the workplace is essential for happiness & your comfort. A lot of office are too hot or cold and it can affect your work so the temp should be medium.

(vi) Location ⇒ One of your major element of productive office spaces is location of your office. It is the place where you will connect your clients, employees and vendors.

(vii) Furniture ⇒ Pure strings, ergonomic furniture is essential for your staff. All desks & chair needs to be fully adjustable, without exception, so that they are comfortable and well-suited to each staff member. better posture → less stress

(viii) Kitchen/Dining space ⇒ basic food essential coofe & tea maker & toaster, microwave, kettle etc. (If there in your budget) to keep your employees hydrated, nourished & energised.

(ix) Electric outlets ⇒ A Room should be analysied first & then at required area electrict outlets should be provied to eleminate unplanned outlets. It can also pose serious health & safety risks. More electric outlets are certainly better than less.

(x) Parking space ⇒ Its very important for every office space. A proper parking should be there for convenience of the Employees.

Commercial Building ⇒ A commercial property is defined as building structures & improvement located on a parcel of commercial real estate intended to generate profit.

* Types :- (i) Industrial (ii) Manufacturing facility (iii) Warehouse (iv) Retail (v) Flex (vi) Shopping mall (vii) Office (viii) Restaurant (ix) Medical (x) Hotel & lodging (xi) Luxury home or Estate (xii) Market & location.

(i) Industrial ⇒ Square footage loading docks for trucks, several HVAC units & several points of electrical distribution, an easily accessible flat roof, & other installed features.

(ii) Manufacturing facility ⇒ This type of bldg is used to produce goods or materials & is easily categorized as either a heavy manufacturing facility or light assembly facility.

- A heavy manufacturing facility tends to make heavy-duty products & has large machinery & equipment. These facilities are typically renovated & customized for specific owners & tenants
- A light assembly facility tends to be smaller & simpler than a heavy manufacturing facility. These facilities also produce smaller goods.

(iii) Warehouse ⇒ is used ~~to~~ for general storage and distribution of goods. open space bldg, where cilings open to the roof's interior structure

~~The layout tends to be an ope~~

- This helps to accomodate high freestanding or installed rack systems.
- A standard office space is divided into seperate rooms, and typically includes restrooms, and a possibly a residential-style-kitchen.

(iv) Retail ⇒ This type of property is where goods or services are sold to customers.

Most Retail spaces have sample parking areas and bordering sidewalks, while some have escalators, elevators, and covered parking structures. [Sub categories :- Shopping centre or Mall, office, Padsite, ...]

(v) Flex ⇒ A bldg that combines more than one usage in single facility is considered a flex commercial property.

[Office Bldg, Medical or Dental office, MDU (Multi-Dwelling Unit), Luxury home, Hotel and Lodging, Restaurant, Park & recreation centre, Market & location..].

Types of Park ⇒
- Community Park
- Mini park
- View park
- Open space
- Public Beach
- School

(i) Commercial Park ⇒ A Commercial park typically refers to an area or facility thats is developed for business purposes. It may include office spaces, retail outlets, parking, swimming pool, facilities for pickniking, active sports and other facilities that serve a larger population.
They are designed or orianted on a particular theme such as active sports or aquatic facilities.

(ii) Mini Park ⇒ are smaller parks which may take one of two different forms. Most mini parks are less than one acre in size, serve a quarter-mile radius, and are located within neighborhood, separate from major or collector roads. * But Sometime serve the entire city and are located as urban trail heads along major trails or streets.

(iii) View Park ⇒ are smaller passive parks designed to take advantage of a significant view. They are often located on costal bluffs to focus upon ocean or bay views. Size ⇒ $1\frac{1}{2} - 3$ Acres & Serve entire City.
~~The~~ View Parks are improved with landscaping, walkways & benches.

(iv) Open Space ⇒ includes passive & active open space acres which do not function as public parks but do provide open space relief.
Accessible to general public.

(v) Public Beach ⇒ Serve a number of local and regional functions.
In some neighbourhood, beaches function as neighbourhood or community parks.

Easy accessible, lack of enterance fees & a lack of other available parks has contributed to this function. Includes active sports, snack bars, showers, drinking fountains, restrooms, walkways, docks, benches, shade trees and parking areas.

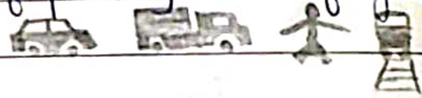
(VI) School :- are a part of the recreation system in the city bcz field and playground areas can serve the general public durking weekends and after school.

Transportation System :- The transportation Element provides policies & actions to maintain & improve the District's transportation system and enhance the travel choices of current of current & future residents, visitors & workers.

* A Transportation system refers to the organized network of infrastructure, vehicles, and facilities that enables the movement of people, goods, or service from one location to another.

Components of Transportation :-

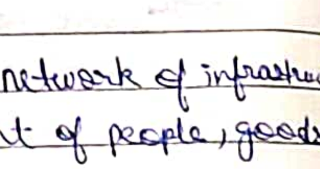
(i) Modes :- • Conveyances used for the mobility of passangers and freight.
• Mobile ~~refers~~ elements of transportation.



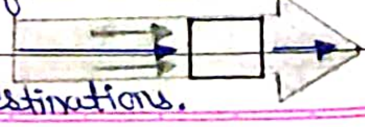
(ii) Infrastructure :- • Physical support of transport modes, such as routes & terminals.

• Fixed elements of transportation including superstructures.

(iii) Networks :- System of linked locations (nodes)
• Functional & spatial organization of transportation.



(iv) Flows :- • Movements of people, freight and information over their network.



• Flows have origins, intermediary locations and destinations.

→ Types of Transportation :-

① **Road Transportation** ⇒ Transportation by road system is the only mode which could given maximum flexibility of service from origin to destination, to one & all.

- Various classes of vehicles such as car, bus, truck, two-wheels etc. may be permitted to make use of the roads.
- Fastest Mode of Transport
- More economical High score of flexibility.
- Less time to Reach destination
- Road transportation develop in different types
  - National Highways (NH)
  - State Highways (SH)
  - Major District Roads (MDR)
  - Other District Roads (ODR)

② **Railways** ⇒ The concept of rail transportation is movement of multiple wagons or a train of wagons or passenger bogies fitted with steel wheels running over two parallel steel rails of the railway track.

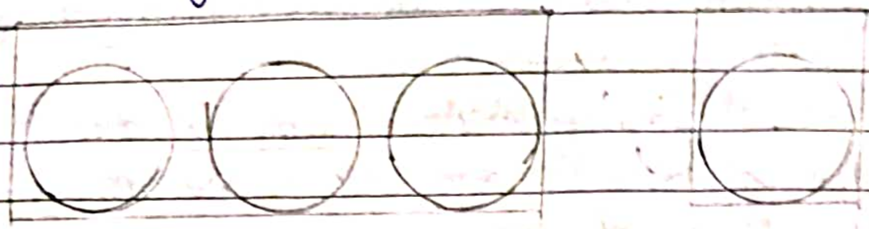
- Can transport large number of passengers or large quantity of goods.
- Used for long distance travel & goods supply.
- Indian Railway is one of the World's largest railway Network.
- Indian Railway were introduced in 1853.

③ **Culverts** ⇒ Culvert is defined as tunnel structure constructed under roadways or Railways to provide cross drainage or take electrical or other cables from one side to other.

- It is totally enclosed by soil or ground.
- Material used for Culvert construction :- Concrete, steel, plastic, Aluminium, High density polythene.

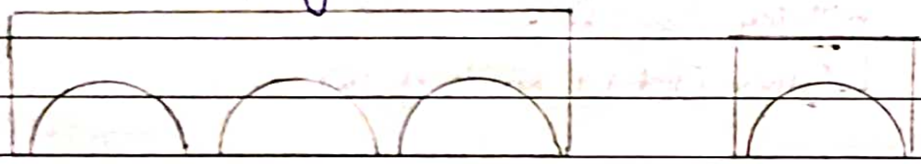
→ Types of Culverts:- Pipe, Box, Arch Culvert etc...

(i) Pipe Culvert (Single or Multiple)



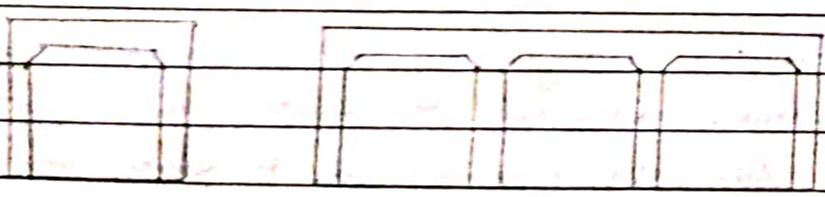
- Rounded in shape
- Single in no or Multiple
- Diameter ranges from 1-6m.
- Made of Concrete or steel.
- They are suitable for large flow very well.

(ii) Pipe Arch Culverts ⇒ (Single or Multiple)



- Half circle shape culvert
- Suitable for large water flow but flow should be stable.
- Can also be provided in multiple no on requirments.
- Becz of arch shape fish or sewage in drainage easily carried.
- Aesthetic View.

(iii) Box Culvert ⇒ (Single or Multiple).

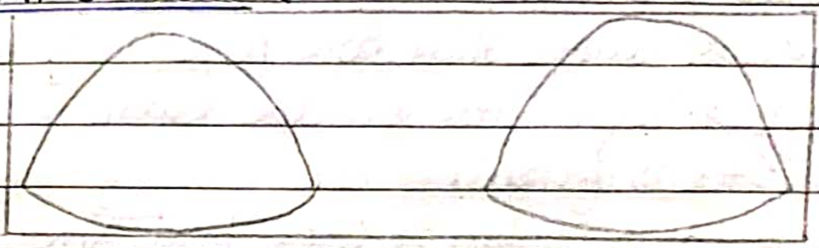


- Rectangular in shape
- Reinforcement is also provided in the Construction of Box Culvert.
- They are used to dispose Rain water. So They are are not ~~use~~ useful

in dry period or climate.

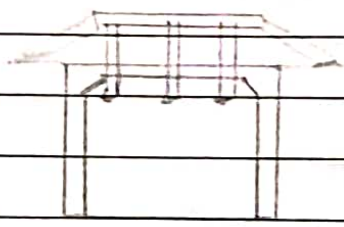
- Also used as passages to cross the rail or Roadway during dry period for animals etc.
- Not suitable for larger velocity becz sharp corners.

(iv) Arch Culvert ⇒



- Similar to pipe arch culvert but in this case an artificial floor is provided below the Arch.
- Used for Narrow passages.
- Made up of Concrete (Artificial floor) & arch also made of Concrete.
- Steel arch culverts are also available but expensive.

(v) Bridge Culvert ⇒



- Similar As Box Culverts.
- Generally these Rectangular shaped culverts can replace the box culvert if artificial floor is not necessary.
- Foundation is laid under ground surface.
- These culverts are provided on Canals or rivers and also used as road bridges for vehicles.

(4) Airways ⇒ Transport by air is the fastest among all 4 modes.

- Air transport provides more comfortable and fast travel Resulting in saving time for the passengers and goods delivery.
- Expensive from all other modes.

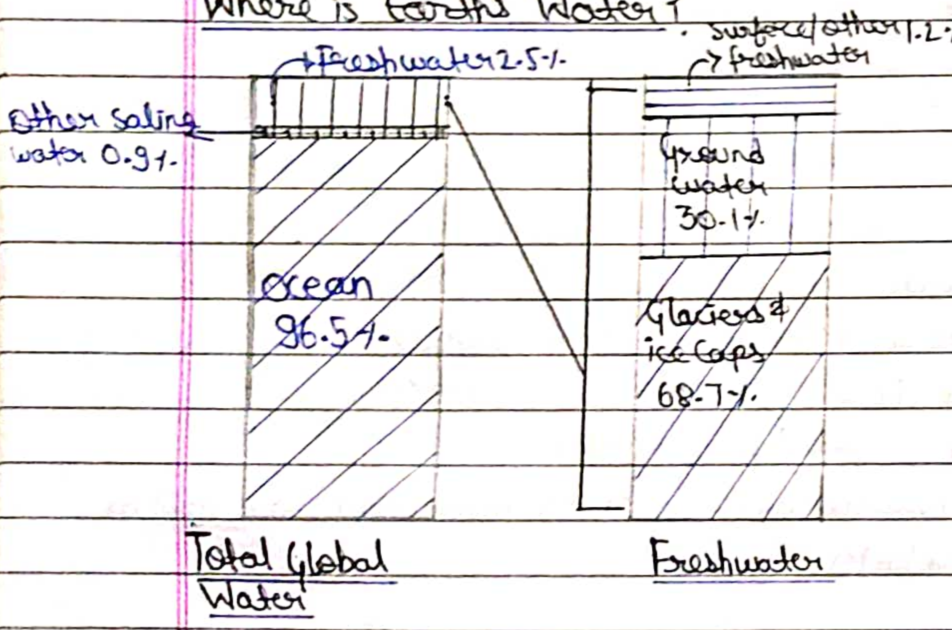
- Airports are ~~data~~ developed to provide facilities for aircraft for take-off, landing, parking and carry out Maintenance work.
- For shorter hauls helicopters are used in ~~heliport~~ helipet are developed for their landing takeoff.

Water Resources & Water System :- Water Resource are Natural Resources of Water that are potentially useful for humans, for ex 97% of water on the earth is salty Water and only 3% Water is fresh :

- Water available in Earth Surface :- Lake, Pond, River, Glaciers, ~~etc~~
- Artificial sources of freshwater can include ~~surface~~ treated waste-water (reclaimed water) and desalinated seawater.

- Uses of water :- Agriculture, Industrial, Household, Recreational and Enviromental activities.

Where is Earths Water?



- Infrastructures :- Is the set of fundamental facilities and systems that support sustainable functional of households & firms.
- Serving a Country, city or other area, including the services &

facilities necessary for its economy to function.

- Infrastructure is composed of public and private physical structures such as.

→ Roads, Bridges, water supply, Electrical grids, Railways, Tunnels, Sewes etc...

Q) Define Infrastructure?

⇒ Infrastructure refers to the fundamental physical & organizational structures & facilities needed for the operation of a society, enterprise, or system. It includes, elements such as roads, bridges, utilities, buildings, and technological network that support and enable various activities & services.

Canals ⇒ Canals are waterways channels, or artificial waterways, for water conveyance, or for servicing water transport vehicles.

- In most cases, a canal has a services of dams and locks that create reservoirs of low speed current flow.
- These reservoirs are referred to as slack water levels, often just called levels.

→ Types of canals :-

(i) Waterways used for carrying vessels transporting goods & people.

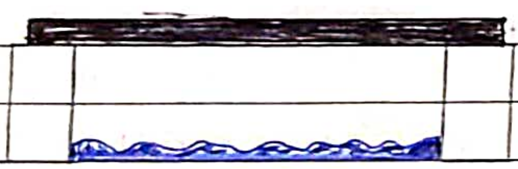
→ feature:- a) Those connecting existing lakes, Rivers, other canals or seas oceans. (b) Those connected city network :- Canal Grande & others Venice. Designed for navigation, irrigation or drainage.

(ii) Aqueducts ⇒ ~~These~~ water supply canals that are used for the conveyance and delivery of portable water for human consumption, municipal uses, hydropower canals and agriculture irrigation.

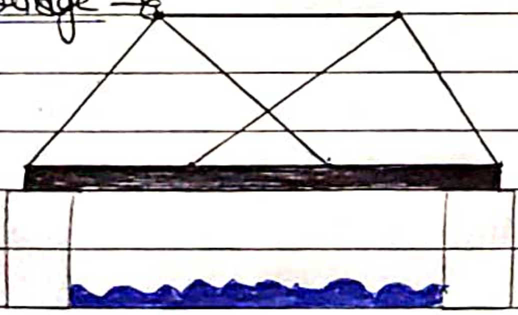
→ **Bridge** :- A Bridge is a structure built ~~to each serving a particular~~ to spam a physical obstacle (such as body of water, Valley, Road & Rail.
There are many diff designs of bridges each serving a particular purpose & applicable to diff situation.

Types of Bridges:-
- Beam bridge
- Truss bridge
- Cantilever bridge
- Arch bridge
- Suspension bridge
- Cable - stayed bridges.

(i) Beam bridge :-



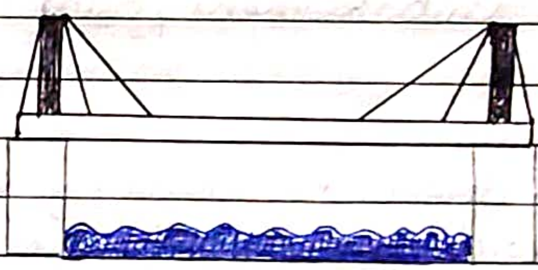
(ii) Truss bridge :-



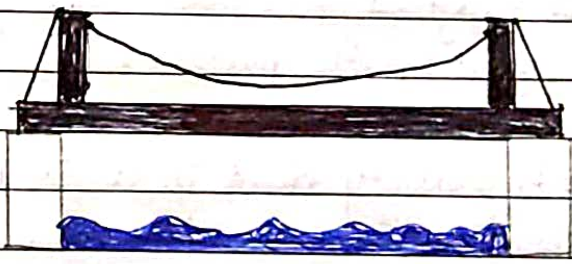
(iii) Cantilever bridge :-

(iv) Arch bridge =>

(v) ~~Suspension~~ Cable-Stayed bridge =>



(vi) Suspension bridge =>



Tunnel => A tunnel is an underground passageway avg through the surrounding soil/earth Rock & enclosed except for entrance & exit commonly at each end.

- A Tunnel may be foot or vehicular Road traffic for rail traffic or for a Canal.
- Some tunnels are used a sewers or adeducts to supply water for hydroelectrode stations. And some are built for, military purposes or by civilian for smuggling of weapons ~~or~~ contraband or people.

Dam => A Dam is a barrier that stops or restructs the flow surface water or underground streams.

- Hydropower is often used in conjuction with dams ~~use~~ to ~~ese~~ generate electricity.
- A dam can also used to collect or store water which can evenly distributed b/w locations

Types:- Archdams, Gravity dams., Arch gravity dams, Burrage dam, Embakement dam, Cofferdams., Natural dams.

(i) Arch Dam => In this dam staibility is obtained by combination of arch and gravity action.

- A arch dam is concrete dam that is curved upstream in plan.
- The arch dam is designed so that the force of water against it, known as hydrostatic pressure.

(ii) Gravity Dam => A gravity dam is a solid structure made of concrete or Masonry constructed area as a river to create reservoir on its upstream.

(iii) Arch Gravity Dam => An arch gravity dam is corporates the arch's curved design which is effective is supporting the water in Narrow.

(iv) Coffer Dams => waterlift enclosure from which water is pumped to expose the of body of water in order to permit construct of pier or other hydraulic work.

(v) Natural Dam => Dam can also be created by Natural geological forces. Most of Natural dams takes formed as a result of landsliding are seen in Mountain areas, with Narrow Valleys & steels slopes.

- Lava dams are formed when lava flows, often basaltic, intercept the path of a stream or lake outlet, resuting in the creation of a natural impoudement.

Ground support system ⇒ Optimising space, achieving suitable alingnments of roads & railways, or safety within an inhabited enviroment are, among many other situations, scenarios that often challange the effect of gravity when it comes to construction, requiring feasible ground support systems.